STAC PS - 3
EXOPLANET HUNTERS

I. PROVIDED DATA

Star	Mass ($M_\odot$)	Radius ($R_\odot$)	Luminosity ($L_\odot$)	Surface Temp (K)
STAC 1	1.00	1.00	1.00	5800
STAC 2	0.80	0.80	0.40	5000
STAC 3	0.50	0.50	0.08	3800

Constants:

- Solar Radius, $R_\odot = 696,000$ km
- Earth Radius, $R_\oplus = 6371$ km
- 1 AU $= 149.6 \times 10^6$ km
- 1 year = 365.25 days
- $M_\odot$ = Mass of the Sun
- $L_\odot$ = Luminosity of the Sun
- 1 AU = Distance between Earth and Sun

II. MISSION

In the rest of the document, STAC X (X = 1, 2, 3) refers to the planet orbiting the STAC X star unless mentioned otherwise.
For recorded values that differ from the usual orbital periods, we assume them to be anomalous readings and do not include them to our calculations.

II..1 ANALYSIS OF TRANSIT DEPTH

Formula:

$$\delta = \frac{F_{\text{normal}} - F_{\text{transit}}}{F_{\text{normal}}}$$

- δ = Transit Depth
- F_{normal} = Brightness outside transit
- F_{transit} = Brightness during transit

STAC 1

$F_{\text{transit}} = 0.997$
$F_{\text{normal}} = 1$
$\delta_{\text{STAC1}} = \dfrac{1 - 0.997}{1}$
$\boxed{\delta_{\text{STAC1}} = 0.003}$

The observed dips occur at days 3, 8, 12, 16 and 20. The intervals are approximately 4 days, indicating a periodic transit.

STAC 2

$F_{\text{transit}} = 0.999$
$F_{\text{normal}} = 1$
$\delta_{\text{STAC2}} = \dfrac{1 - 0.999}{1}$
$\boxed{\delta_{\text{STAC2}} = 0.001}$

The observed dips occur at days 20, 100 and 180, indicating a periodic transit every 80 days.

STAC 3 The regularly spaced dips occur at days 20, 60, 100 and 140. We use these four observations to calculate the average transit brightness.

$$F_{\text{transit}} = \frac{0.9990 + 0.9992 + 0.9991 + 0.9993}{4}$$
$F_{\text{transit}} = 0.99915$
$F_{\text{normal}} = 1$
$\delta_{\text{STAC3}} = \dfrac{1 - 0.99915}{1}$
$\boxed{\delta_{\text{STAC3}} = 0.00085}$

II..2 DETERMINATION OF PLANET SIZE

Formula:

$$\delta \approx \left(\frac{R_p}{R_s}\right)^2$$

Rearranging:

$$R_p = R_s\sqrt{\delta}$$

- R_p = Radius of the Planet
- R_s = Radius of the Star
- δ = Transit Depth
- For Reference (Radii):
  - Jupiter: 69,911 km
  - Neptune: 24,764 km
  - Super-Earth: approximately 8,000 km to 12,750 km
  - Earth: 6,371 km

STAC 1

$$R_p = 1 \times 696000 \times \sqrt{0.003}$$
$$R_p \approx 38121.49 \text{ km}$$
$$R_p = \frac{38121.49}{6371}$$
$$\boxed{R_p \approx 5.98 R_\oplus}$$

Category = Large Neptune / Sub-Saturn Sized

STAC 2

$$R_p = 0.8 \times 696000 \times \sqrt{0.001}$$
$$R_p \approx 17607.56 \text{ km}$$
$$R_p = \frac{17607.56}{6371}$$
$$\boxed{R_p \approx 2.76 R_\oplus}$$

Category = Sub-Neptune Sized

STAC 3

$$R_p = 0.5 \times 696000 \times \sqrt{0.00085}$$
$$R_p \approx 10145.86 \text{ km}$$
$$R_p = \frac{10145.86}{6371}$$
$$\boxed{R_p \approx 1.59 R_\oplus}$$

Category = Super-Earth Sized

II..3 DETERMINATION OF PLANETARY ORBITS

Formula:

$$P^2 = \frac{a^3}{M}$$

Rearranging:

$$a = \sqrt[3]{MP^2}$$

- a = Orbital Distance (AU)
- P = Orbital Period (Years)
- M = Mass of the Star (Solar Masses)
- 1 year = 365.25 days
- We assume observations that significantly deviate from the periodic pattern to be anomalous and omit them from orbital-period calculations.

STAC 1

$$P_{\text{orbital}} = 4 \text{ days}$$
$$a = \left(\left(\frac{4}{365.25} \right)^2 \times 1.00 \right)^{1/3}$$
$$\boxed{a \approx 0.04932 \text{ AU}}$$
$$a = 0.0493151 \times 149.6 \times 10^6$$
$$\boxed{a \approx 7,377,538 \text{ km}}$$

STAC 2

$$P_{\text{orbital}} = 80 \text{ days}$$
$$a = \left(\left(\frac{80}{365.25} \right)^2 \times 0.80 \right)^{1/3}$$
$$\boxed{a \approx 0.33731 \text{ AU}}$$
$$a = 0.3373105 \times 149.6 \times 10^6$$
$$\boxed{a \approx 50,461,648 \text{ km}}$$

STAC 3

$$P_{\text{orbital}} = 40 \text{ days}$$
$$a = \left(\left(\frac{40}{365.25} \right)^2 \times 0.50 \right)^{1/3}$$
$$\boxed{a \approx 0.18168 \text{ AU}}$$
$$a = 0.1816784 \times 149.6 \times 10^6$$
$$\boxed{a \approx 27,179,081 \text{ km}}$$

All three planets orbit significantly closer to their host stars than Earth does to the Sun.

II..4 DETERMINATION OF MOST PROMISING PLANET

Formula:

$$d_{\text{HZ}} \approx \sqrt{\frac{L}{L_\odot}} \text{ AU}$$

- d_{HZ} = Approximate reference distance of the habitable zone
- L = Luminosity of the Star
- $L_\odot$ = Luminosity of the Sun

The difference between the planet's orbital distance and the estimated habitable-zone reference distance is:

$$\Delta d = |d_{\text{HZ}} - a|$$

STAC 1

$d_{\text{HZ}} = \sqrt{1.00}$
$d_{\text{HZ}} = 1.00$ AU
$\Delta d = |1.00 - 0.04932|$
$\boxed{\Delta d \approx 0.95068 \text{ AU}}$

STAC 2

$d_{\text{HZ}} = \sqrt{0.40}$
$d_{\text{HZ}} \approx 0.63246$ AU
$\Delta d = |0.63246 - 0.33731|$
$\boxed{\Delta d \approx 0.29515 \text{ AU}}$

STAC 3

$d_{\text{HZ}} = \sqrt{0.08}$
$d_{\text{HZ}} \approx 0.28284$ AU
$\Delta d = |0.28284 - 0.18168|$
$\boxed{\Delta d \approx 0.10116 \text{ AU}}$

STAC 3 is the closest to its estimated habitable-zone reference distance. Its Super-Earth size also makes it an interesting candidate for further investigation.

However, this simplified formula does not calculate the actual inner and outer boundaries of the habitable zone. Therefore, it cannot independently establish whether a planet is inside the habitable zone.

II..5 DETERMINATION OF PLANET EQUILIBRIUM TEMPERATURE

Formula:

$$T_{\text{eq}} = T_s \times \left(\frac{R_s}{2d}\right)^{1/2} \times (1 - A)^{1/4}$$

- T_{eq} = Planetary Equilibrium Temperature
- T_s = Surface Temperature of the Star
- R_s = Radius of the Star
- d = Orbital Distance (in km)
- A = Bond Albedo (Reflectivity of the Planet)
- Since no albedo values are provided, we assume an Earth-like Bond Albedo of 0.3.

STAC 1

$$T_{\text{eq}} = 5800 \times \left(\frac{696000}{2 \times 7377538}\right)^{1/2} \times (1 - 0.3)^{1/4}$$

$\boxed{T_{\text{eq}} \approx 1152 \text{ K}}$
$T_{\text{eq}} = 1152.22 - 273.15$
$\boxed{T_{\text{eq}} \approx 879^\circ\text{C}}$

STAC 2

$$T_{\text{eq}} = 5000 \times \left(\frac{556800}{2 \times 50461648}\right)^{1/2} \times (1 - 0.3)^{1/4}$$

$\boxed{T_{\text{eq}} \approx 340 \text{ K}}$
$T_{\text{eq}} = 339.70 - 273.15$
$\boxed{T_{\text{eq}} \approx 67^\circ\text{C}}$

STAC 3

$$T_{\text{eq}} = 3800 \times \left(\frac{348000}{2 \times 27179081}\right)^{1/2} \times (1 - 0.3)^{1/4}$$

$\boxed{T_{\text{eq}} \approx 278 \text{ K}}$
$T_{\text{eq}} = 278.11 - 273.15$
$\boxed{T_{\text{eq}} \approx 5^\circ\text{C}}$

III. FINAL CONCLUSION

Planet	Distance to HZ Reference	Equilibrium Temp	Temp (°C)
STAC 1	0.95068 AU	1152 K	879°C
STAC 2	0.29515 AU	340 K	67°C
STAC 3	0.10116 AU	278 K	5°C

Table 1: Comparison of the Three Planets

Based on the calculated equilibrium temperatures:

- **STAC 1:** Extremely hot ($\sim$ 879°C), making it a very unlikely candidate for liquid surface water.
- **STAC 2:** Relatively warm ($\sim$ 67°C). Its habitability would depend heavily on atmospheric conditions and heat distribution.
- **STAC 3:** Relatively cool ($\sim$ 5°C), making it the most promising candidate. A suitable atmosphere and greenhouse effect could help maintain conditions favourable for liquid water.

Since Princess Leia and the Rebels are humans (excluding the droids, who can survive in a wider range of environments), the fleet should hyperdrive towards **STAC 3**. Its Super-Earth-sized planet and relatively favourable estimated

equilibrium temperature make it the strongest candidate for further habitability investigations.

An additional observation that could help determine its suitability is atmospheric spectroscopy. This would help identify gases such as water vapour, carbon dioxide, oxygen and methane, providing information about its atmosphere and possible habitability.

P.S -

- Equilibrium temperatures are approximate values and do not necessarily represent actual surface temperatures. Atmospheric composition, greenhouse effects, clouds and planetary rotation can significantly affect surface conditions.